

- 2.14 understand the importance in plants of turgid cells as a means of support**
- 2.15 understand the factors that affect the rate of movement of substances into and out of cells, to include the effects of surface area to volume ratio, temperature and concentration gradient
- 2.16 describe experiments to investigate diffusion and osmosis using living and non-living systems.

e) Nutrition

Students will be assessed on their ability to:

Flowering plants

- 2.17 describe the process of photosynthesis and understand its importance in the conversion of light energy to chemical energy
- 2.18 write the word equation and the balanced chemical symbol equation for photosynthesis
- 2.19 understand how varying carbon dioxide concentration, light intensity and temperature affect the rate of photosynthesis
- 2.20 describe the structure of the leaf and explain how it is adapted for photosynthesis
- 2.21 understand that plants require mineral ions for growth and that magnesium ions are needed for chlorophyll and nitrate ions are needed for amino acids
- 2.22 describe experiments to investigate photosynthesis, showing the evolution of oxygen from a water plant, the production of starch and the requirements of light, carbon dioxide and chlorophyll

Humans